OPTIMIZATION PROJECT REPORT

A Quantitative Approach to Cricket Team Selection

Squad and MI Playing XI Optimization

TEAM MEMBERS

PRATEEK N JAISWAL	*BSD-BG-2412*
MUKESH MURUGALINGAM	*BSD-BG-2409*
ANANT AGARWAL	*BSD-BG-2402*

Project Semester II - 2025

Contents

1 Problem Statement **2**
1.1 Research Questions 2

2 Plan **2**
2.1 IPL Squad Optimization Strategy 2
2.2 MI Playing XI: Success Score (S_i) 2

3 Analysis **4**
3.1 Auction Pool Distribution 4
3.2 MI Playing XI Optimization 4

4 Mathematical Formulation **5**
4.1 IPL Squad Optimization (Part A) 5
4.2 MI Playing XI Optimization (Part B) 6

5 Results and Conclusions **7**
5.1 IPL Squad Optimization 7
5.2 MI Playing XI Optimization 7
5.3 Summary 7
5.4 Discussion and Limitations 7

1 Problem Statement

This report presents a double optimization strategy for the Mumbai Indians (MI) franchise. The objective is to use Linear Programming (LP) and performance analytics to model the "Auction Paradox" the challenge of balancing a limited budget with high-performance team.

- **IPL Squad Optimization:** Selection of a 23-player squad from the auction pool to maximize overall "Impact Potential".
- **MI Playing XI Optimization:** Selection of the best 11 players from the existing squad to maximize expected performance score per match.

1.1 Research Questions

1. How can we mathematically maximize team strength within rigid auction subgroup constraints?
2. What is the statistically "best" Playing XI for MI based on a weighted 3-year performance index?

2 Plan

2.1 IPL Squad Optimization Strategy

The squad selection is modeled as a Linear Programming (LP) problem. Players are categorized into subgroups (Domestic, International, All-rounders) with assigned weights based on historical league impact.

- Assign weights to players based on their subgroups (Domestic / Overseas, Batsman, Bowlers, All-rounders).
- Define a Linear Programming problem to maximize squad performance.
- Apply constraints on subgroup sizes and total squad size.

  Player Groups:

  - Domestic Batsmen, Bowlers, All-rounders
  - Indian International Batsmen, Bowlers, All-rounders
  - Overseas International Batsmen, Bowlers, All-rounders

  Source: TATA IPL 2025 Auction Pool

2.2 MI Playing XI: Success Score (S_i)

- **What to Measure?**
- Success Score (random variable) for each player based on last 3-year performance:
  - Batting: runs, strike rate, average

  - Bowling: wickets, economy, average
  - All-round: composite of both
  - Role: right/left batsman, off/leg spinner, pacer, keeper, all-rounder

- **How?**
  1. Convert performance statistics into a Success Score $S_i \in [0, 1]$.
  2. Treat S_i as the expected performance (random variable).
  3. Build the optimization problem with:
     - Objective: Maximize total expected score
     - Constraints: Playing XI composition rules (e.g., maximum 4 foreign players)
     - Success Score formulation as defined above

Success Score Formulations

$$\textbf{Batsmen:}\quad S_b = 0.4\left(\frac{\text{Batting Avg}}{50}\right) + 0.4\left(\frac{\text{SR}}{150}\right) + 0.2\left(\frac{\text{Innings} > 30}{\text{Total Innings}}\right)$$

$$\textbf{Bowlers:}\quad S_{bowl} = 0.3\left(\frac{6}{\text{Economy}}\right) + 0.4\left(\frac{25}{\text{SR}}\right) + 0.3\left(\frac{\text{Wkts/Match}}{3}\right)$$

$$\textbf{All-rounders:}\quad S_{ar} = 0.5 \cdot S_b + 0.5 \cdot S_{bowl}$$

$$\textbf{Keepers:}\quad S_{wk} = 0.7 \cdot S_b + 0.3\left(\frac{\text{Dismissals}}{\text{Matches}}\right)$$

- We use data gathered from the IPL official website.
- The dataset consists of player statistics collected from previous IPL records.
- **Source:** Mumbai Indians Official Squad

name	role	BatAvg	SR	30s	Matches	Econ	BowlSR	WktsperMatch	Dismissals	Si(success score)	Foreign
Rohit Sharma	RHB	23.65	142.22	13	44	0.0	0.0	0.0	0.0	0.628	0
Tilak Varma	LHB	39.86	145.54	17	38	0.0	0.0	0.0	0.0	0.862	0
Suryakumar Yadav	RHB	39.19	166.68	15	35	0.0	0.0	0.0	0.0	0.883	0
Bevon Jacobs	RHB	30.66	180.30	1	6	0.0	0.0	0.0	0.0	0.759	1
Ryan Rickelton	LHB,WK	42.25	156.12	7	17	0.0	0.0	0.0	0.0	0.797	1
Hardik Pandya	RHB,AR,RHB	26.14	136.54	10	45	9.1	24.95	0.5	0.0	45	0
Mitchell Santner	LHB,AR,OS	7.6	109	0.0	12	6.7	26.67	0.75	0.0	12	1
Karn Sharma	LS	7	120.27	0.0	21	8.39	19.23	1	0.0	21	0
Trent Boult	LHP,P	14	104.94	0	42	8.34	20.44	1.07	-	42	0
Jasprit Bumrah	RHP,P	9.33	87.34	0.0	41	7.05	18.71	1.36	0.0	41	1
Deepak Chahar	RHB,P	1	144	0.0	33	8.14	19.23	1	0.0	33	0
Will Jack	RHB,OS	32.86	175.57	1	8	11.11	27	0.25	0.0	8	0
Arjun Tendulkar	LHB,P	13	144.44	0.0	4	9.36	19.67	0.75	0.0	4	1
Naman Dhir	RHB	23.33	177.22	2	7	0.0	0.0	0.0	0.0	7	0
Reece Topley	LHP,P	0.0	0.0	0.0	1	7	20	1	0.0	1	1
Raj Angad B	LHB,RHP,AR	5.5	78.57	0.0	2	0.0	0.0	0.0	0.0	2	0

Table 1: Player statistics from MI data

3 Analysis

3.1 Auction Pool Distribution

IPL Squad Optimization

Price_in_Lacs	Number_of_Players	Impact_Score
200	82	10
150	27	9
125	18	9
100	23	9
75	92	8
50	8	8
40	5	6.5
30	20	6.5

Price_in_Lacs	Number_of_Players	Avg_Probability_Impact
200	82	0.15
150	27	0.13
125	18	0.13
100	23	0.13
75	92	0.12
50	8	0.12
40	5	0.09
30	20	0.09

3.2 MI Playing XI Optimization

We assign players into IID categories using role tags (e.g., RHB, LHB, Pacer, Off-spin):

- Batting statistics
- Bowling statistics
- Normalized score between 0–1
- $x_i \in \{0, 1\}$: whether to pick player i (1 if selected, 0 otherwise)

Player Role Sets:

$$\begin{aligned}
\text{LHB} &= \{\text{Tilak V}, \text{Robin M}, \text{Rickelton}\} \\
\text{RHB} &= \{\text{Rohit S}, \text{SKY}, \text{Bevon J}, \text{Shrijith K}\} \\
\text{Pr} &= \{\text{T Boult}, \text{D Chahar}, \text{Ashwani K}, \text{R Topley}, \text{Satyanarayana P}, \text{Arjun T}, \text{J Bumrah}\} \\
\text{S} &= \{\text{Karan S}, \text{Mujeeb R}\} \\
\text{ARP} &= \{\text{Hardik P}, \text{W Jacks}, \text{R Angad B}, \text{C Bosch}\} \\
\text{AROS} &= \{\text{V Puthur}, \text{M Santner}, \text{Naman D}\} \\
\text{WK} &= \{\text{Robin M}, \text{R Rickelton}, \text{Shrijith K}\}
\end{aligned}$$

where RHB denotes right-handed batsmen, LHB denotes left-handed batsmen, ARP denotes all-rounder pacers, AROS denotes all-rounder spinners, and WK denotes wicket keepers.

Variables:

- Player name
- Role (e.g., LHB, Pacer, WK, AR)
- Nationality (Indian/foreign)
- Batting average and strike rate
- Bowling economy, strike rate, and wickets
- Dismissals (for wicket keepers)

The success score is computed as a weighted sum of normalized variables. Missing values are imputed using mean substitution.

4 Mathematical Formulation

4.1 IPL Squad Optimization (Part A)

Objective: $\max Z = 90x_1 + 26x_2 + 51x_3 + 80x_4$
Subject to:

$$4 \leq x_1 \leq 8 \quad \text{(Batsmen constraints)} \tag{1}$$
$$2 \leq x_2 \leq 6 \quad \text{(Spinners constraints)} \tag{2}$$
$$2 \leq x_3 \leq 6 \quad \text{(Pacers constraints)} \tag{3}$$
$$5 \leq x_4 \leq 9 \quad \text{(All-rounders constraints)} \tag{4}$$
$$\sum x_i = 23 \quad \text{(Total Squad Size)} \tag{5}$$

$$x_1, x_2, x_3, x_4 \in \mathbb{Z}_{\geq 0}$$

Decision Variables and Weights:

$$\begin{aligned} x_1 &= \text{Number of batsmen}, \quad w_1 = 90 \\ x_2 &= \text{Number of spin bowlers}, \quad w_2 = 26 \\ x_3 &= \text{Number of fast bowlers}, \quad w_3 = 51 \\ x_4 &= \text{Number of all-rounders}, \quad w_4 = 80 \end{aligned}$$

- We used maximum and minimum limit we used here for each sub-group's, is the supremum and infimum of players for the same sub-groups for every IPL Teams of this year.
- We assigned the average probability of impact of each sub-group (assuming them to be identically Independent for each sub-group with different probability for each subgroup) of players values between [0,1] and used it to assign the average weights [impact of each sub-group on squads performance] we also randomised it by adding values [1/2/3/4] based on which type[fast/spin/batsman/all-round] is the player.
- Similarly we calculated the maximum scores for each sub-group and every team then we used it's supremum to assign it the each sub-groups total.

- We limit the total players to be 23 just to make our next part more efficient.
- We used one assumption here as players with base price above 50L are 90% likely to be sold we did not change the data in this case, but for players below 50L only 20 players on average gets selected so we reduced the data to 20.

4.2 MI Playing XI Optimization (Part B)

Total Players: 23

Role Breakdown: 4 batsmen (1 LHB), 7 pacers (2 LHF), 2 spinners (OS, 1 LS), 4 all-rounder pacers (1 LHB), 3 all-rounder spinners (1 LHS), 3 wicketkeepers (2 LHB)

Decision Variables:

- $x_i \in \{0, 1\}$: decision variable indicating whether player i is selected
- S_i: success score of player i

Objective Function:

$$\max \sum_{i=1}^{23} S_i x_i$$

This is a linear (and convex) objective.

Constraints:

$$\begin{aligned}
\text{Total players:} \quad & \sum_{i=1}^{23} x_i = 11, & \text{Leg spinner:} \quad & \sum_{i \in LS} x_i \geq 1 \\
\text{Foreign players:} \quad & \sum_{i \in F} x_i \leq 4, & \text{Wicketkeepers:} \quad & \sum_{i \in WK} x_i \geq 1 \\
\text{Pacers:} \quad & \sum_{i \in P} x_i \geq 3, & \text{Left-handed batsmen (LHB):} \quad & \sum_{i \in LHB} x_i \leq 2 \\
\text{Left-handed pacer (LHP):} \quad & \sum_{i \in LHP} x_i \geq 1, & \text{Right-handed batsmen (RHB):} \quad & \sum_{i \in RHB} x_i \geq 2 \\
\text{Off-spin all-rounder:} \quad & \sum_{i \in OSA} x_i \geq 1, & \text{Pacer all-rounders:} \quad & \sum_{i \in ARP} x_i \geq 1
\end{aligned}$$

$$x_i \in \{0, 1\}, \quad i = 1, 2, \ldots, 23$$

Code: The implementation can be found in the GitHub repository: optimization-project-sem-2

5 Results and Conclusions

5.1 IPL Squad Optimization

- 2 Indian Domestic Batsmen
- 3 Indian International Batsmen
- 3 Overseas International Batsmen
- 0 Indian Domestic Spin Bowlers
- 1 Indian International Spin Bowler
- 1 Overseas International Spin Bowler
- 1 Indian Domestic Fast Bowler
- 2 Indian International Fast Bowlers
- 1 Overseas International Fast Bowler
- 2 Indian Domestic All-rounders
- 4 Indian International All-rounders
- 3 Overseas International All-rounders

5.2 MI Playing XI Optimization

- Rohit Sharma
- Tilak Varma
- Suryakumar Yadav
- Bevon Jacobs
- Ryan Rickelton
- Hardik Pandya
- Mitchell Santner
- Karn Sharma
- Trent Boult
- Jasprit Bumrah
- Deepak Chahar

5.3 Summary

We successfully formulated and solved two optimization problems for cricket team selection:

- **IPL Squad Optimization:** Achieved an optimal squad composition maximizing overall impact.
- **MI Playing XI Optimization:** Selected the best playing XI maximizing historical success scores.

5.4 Discussion and Limitations

Both problems demonstrate how linear programming can guide decision making in sports management.

Limitations:

1. The model assumes that player performances are independent and identically distributed (IID) across subgroups.
2. The success score (S_i) is a simplified, linear function of a few performance metrics. This may overlook contextual factors such as match conditions, opposition strength, or current form.
3. Player roles are tagged simplistically (e.g., LHB, RHB, AR), whereas many players are multidimensional. The model may not fully utilize such versatility.

Contribution:

- **Part A of the project:** Prateek
- **Part B of the project:** Mukesh
- **Code implementation, debugging, data collection of Part A, and report writing:** Anant